

Noisy calibration, latent coverage and the location of the mean

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SUMMARY

Prediction intervals for a latent quantity, such as the mean of an unsampled area, are often calibrated on noisy proxies with known Gaussian error variances. Symmetry of the latent distribution makes such calibration conservative, but symmetry is seldom plausible. Over all log-concave latent distributions, the smallest latent radius that a noisy coverage level guarantees is governed by the distance between the latent mean and the boundary of the calibrated interval. If that distance is bounded away from zero, noisy calibration undercovers by at most an amount of the order of the noise variance, for every noise level, and asymmetric distributions with mean zero show that this order is attained. Over all log-concave distributions the shortfall is of the order of the noise standard deviation, with an explicit sharp constant; this slower rate occurs exactly when the mean lies within a bounded multiple of the noise standard deviation of the boundary, and the limiting coefficient is then a one-dimensional extremum. A calibration slack of one part in a thousand at the ninety per cent level removes the shortfall at every noise level. With heterogeneous noise this gives a finite-sample procedure whose conditional reliability holds for every log-concave latent distribution.

Some key words: Conformal prediction; Deconvolution; Log-concavity; Measurement error; Small area estimation.

1. INTRODUCTION

Suppose that a residual W , the latent target minus a centre fitted on independent training data, is observed only through $V = W + e$, where $e \sim N(0, D)$ is independent of W and D is known. A survey statistician who calibrates a prediction interval for the true mean of an unsampled area on the direct estimates of sampled areas is in this position (Fay & Herriot, 1979), as is a conformal analyst whose calibration responses carry measurement error (Einbinder et al., 2024; Cohen et al., 2025). The threshold t at which the noisy scores $|V|$ have coverage p is observable. We ask which radius gives the latent residual coverage q , uniformly over a class of latent distributions.

If W is symmetric and unimodal, the theorem of Anderson (1955) gives $\text{pr}(|W + e| \leq t) \leq \text{pr}(|W| \leq t)$, so the noisy threshold is conservative (Einbinder et al., 2024). Without symmetry the available transfers make no shape assumption: a median-zero noise turns a noisy level $1 - \alpha$ into a latent level $1 - 2\alpha$, with high-probability versions under a Beta calibration (Guille-Escuret & Ndiaye, 2024), and LatentCP (Zheng et al., 2026) uses a Markov-type compatibility threshold. Cohen et al. (2025) estimate the noise-free threshold by deconvolution without a finite-sample latent guarantee, and Armstrong et al. (2022) give worst-case empirical Bayes intervals for sampled units under moment constraints. For a fixed smooth latent distribution, Gaussian noise moves quantiles by $O(D)$ (Jochmans & Weidner, 2024), which suggests that the error of noisy calibration is of second order.

Uniformly over a shape class the picture is different, and it is organized by the location of the mean. We work with the class $\mathcal{L}$ of log-concave laws on the real line, including point masses, which is closed under convolution, conditioning on intervals, translation and scaling, and contains the usual models for regression residuals. After a reduction of the sharp radius to a two-parameter family (§ 2), we show that without a restriction on the mean the worst-case shortfall at equal levels is of order $D^{1/2}$, with a sharp constant (Theorem 1); that if the mean lies at distance d from the boundary of the interval the shortfall is at most $D/(2d)$ for every D , an order attained by asymmetric laws with mean zero (Theorem 2, Proposition 2); and that the order $D^{1/2}$ occurs exactly when the mean lies within $O(D^{1/2})$ of the boundary, with a limiting coefficient given by a one-dimensional extremum (Theorem 3). A calibration slack of order 10^{-3} removes the shortfall at every noise level, and with heterogeneous known noise this yields a finite-sample procedure (§ 4).

2. THE SHARP RADIUS AND ITS REDUCTION

After scaling t to 1, write $x = D/t^2$ and let $Z \sim N(0, 1)$ be independent of W . Define

$$R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{L}, \text{pr}(|W + x^{1/2}Z| \leq 1) \geq p\}, \quad (1)$$

where Q_q is the lower q -quantile, and $R_{p,q}(x) = -\infty$ if nothing is feasible, which happens exactly when $\text{pr}(|x^{1/2}Z| \leq 1) < p$. If calibration certifies noisy coverage at least p at threshold t , then $t R_{p,q}(D/t^2)$ is the smallest radius that guarantees latent coverage q for every log-concave latent law. This is sharpness within the class and for one constraint, not optimality among procedures that use the whole calibration sample. For $0 \leq b < 1$ let $R_{p,q}^{(b)}(x)$ be defined by (1) with the additional constraint $|E(W)| \leq b$, so that $1 - b$ bounds the distance of the mean from the boundary.

Write $g_x(w) = \text{pr}(|w + x^{1/2}Z| \leq 1) = \Phi\{(1 - w)/x^{1/2}\} - \Phi\{(-1 - w)/x^{1/2}\}$, so that the constraint in (1) reads $E\{g_x(W)\} \geq p$. Let $\mathcal{F}$ be the family of point masses and of laws with density proportional to $e^{\beta w}$ on a segment $[a, b]$.

PROPOSITION 1. $R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{F}, E\{g_x(W)\} \geq p\}$. The same holds for any continuous constraint kernel that vanishes at $\pm\infty$, in particular for the average kernel of § 4.3.

The proof, in the Supplementary Material, truncates W to an interval outside which the kernel is below p , which keeps the constraint, and applies the localization theorem of Fradelizi & Guédon (2004). For $\beta > 0$ and $\sigma = x^{1/2}$, integration by parts gives

$$\int_a^b e^{\beta w} \Phi\left(\frac{c - w}{\sigma}\right) dw = \frac{1}{\beta} \left[e^{\beta w} \Phi\left(\frac{c - w}{\sigma}\right) \right]_a^b + \frac{e^{\beta c + \beta^2 \sigma^2 / 2}}{\beta} \left\{ \Phi\left(\frac{b - c - \beta \sigma^2}{\sigma}\right) - \Phi\left(\frac{a - c - \beta \sigma^2}{\sigma}\right) \right\}, \quad (2)$$

so the constraint is a combination of normal distribution functions. For fixed slope and length, $c \mapsto \text{pr}(|W + c + e| \leq 1)$ and $c \mapsto \text{pr}(|W + c| \leq s)$ are log-concave (Prékopa, 1973), so the feasible translations form an interval and the objective is maximized at one of its ends. The supremum in (1) is therefore a two-dimensional maximization of closed-form quantities.

3. WORST-CASE SHORTFALL AND THE LOCATION OF THE MEAN

3.1. A one-sided problem

Near each end of the interval the problem is one-sided. For $u > 0$ let E_u be exponential with rate u , independent of Z , and for the exponential tail $b - E_u$ let

$$m_u(b) = \text{pr}(b - E_u + Z \leq 0) = \Phi(-b) + e^{-ub + u^2/2} \Phi(b - u),$$

which is continuous and strictly decreasing in b . Let $b_u(p)$ solve $m_u(b) = p$, and write $v_q(u) = b_u(q) - u^{-1} \log(1/q)$ and $k_q(u) = u^{-1} - b_u(q)$. The law $b_u(q) - E_u$ has q -quantile $v_q(u)$ and mean $-k_q(u)$. 75

LEMMA 1. Let $c_{p,q} = \sup\{F_Y^{-1}(q) : Y \in \mathcal{L}, \text{pr}(Y + Z \leq 0) \geq p\}$ and $c_q = c_{q,q}$. Then $c_{p,q} = \sup_{u>0}\{b_u(p) - u^{-1} \log(1/q)\}$, $c_q > 0$, and $q \mapsto c_q$ is nonincreasing.

LEMMA 2. Let $e^{-1} < q < 1$ and $\kappa \geq 0$. Then 80

$$\sup\{F_Y^{-1}(q) : Y \in \mathcal{L}, E(Y) \leq -\kappa, \text{pr}(Y + Z \leq 0) \geq q\} = L_q(\kappa) = \sup\{v_q(u) : u > 0, k_q(u) \geq \kappa\}.$$

The function L_q is positive and nonincreasing, $L_q(\kappa) = c_q$ for $\kappa \leq \kappa_q^*$, where κ_q^* is the largest value of k_q at a maximizer of v_q , and $\kappa L_q(\kappa) \rightarrow \{1 - \log(1/q)\}/2$ as $\kappa \rightarrow \infty$.

Both proofs replace Y by the exponential tail through a supergradient of $\log F_Y$ at its q -quantile; the replacement lies stochastically below Y , so it keeps the quantile, the noisy constraint and the bound on the mean. Ball arithmetic gives $c_{0.8} \in [0.0460730, 0.0460731]$, $c_{0.9} \in [0.0190618, 0.0190619]$, $c_{0.95} \in [0.0084075, 0.0084077]$ and $c_{0.99} \in [0.0013950, 0.0013958]$. In double precision, $\kappa_q^* = 6.81, 20.18$ and 50.07 for $q = 0.8, 0.9$ and 0.95 . 85

3.2. No restriction on the mean

THEOREM 1. For every $q \in (0, 1)$ and every $x > 0$, $R_{q,q}(x) \leq 1 + c_q x^{1/2}$. Moreover $R_{q,q}(x) = 1 + c_q x^{1/2} + o(x^{1/2})$ as $x \rightarrow 0$. 90

Proof of the upper bound. Let $W \in \mathcal{L}$ with $\text{pr}(|W + x^{1/2}Z| \leq 1) \geq q$, and let $\alpha_R = \text{pr}(W + x^{1/2}Z > 1)$ and $\alpha_L = \text{pr}(W + x^{1/2}Z < -1)$, so that $\alpha_L + \alpha_R \leq 1 - q$. The law of $Y = (W - 1)/x^{1/2}$ is log-concave and $\text{pr}(Y + Z \leq 0) = 1 - \alpha_R$. By Lemma 1, $F_Y^{-1}(1 - \alpha_R) \leq c_{1-\alpha_R} \leq c_q$, that is, $\text{pr}(W > 1 + c_q x^{1/2}) \leq \alpha_R$. The same argument applied to $-W$ gives $\text{pr}(W < -1 - c_q x^{1/2}) \leq \alpha_L$, so $\text{pr}(|W| \leq 1 + c_q x^{1/2}) \geq q$. 95

The lower bound, proved in the Supplementary Material, takes $W = 1 + x^{1/2}\{b_u(q + \eta_x) - E_u\}$ at a near-optimal rate u , where $\eta_x \rightarrow 0$ compensates the exponentially small mass that leaks through the far end. Its density rises exponentially to a cut-off just outside the threshold, within a boundary layer of width $x^{1/2}$, and noise draws more mass inwards across that end than it pushes out. Its mean, $1 - k_q(u)x^{1/2}$, is within $O(x^{1/2})$ of the boundary; the next results show that this is necessary. 100

3.3. The mean away from the boundary

THEOREM 2. Let $W \in \mathcal{L}$ with $|E(W)| \leq B$, let $\sigma > 0$ and $t > B$, and suppose that $\text{pr}(|W + \sigma Z| \leq t) \geq q$. Then

$$Q_q(|W|) \leq B + \{(t - B)^2 + \sigma^2\}^{1/2} \leq t + \frac{\sigma^2}{2(t - B)}.$$

In particular $R_{p,q}^{(b)}(x) \leq 1 + x/\{2(1 - b)\}$ for all $p \geq q$ and $x > 0$, and $Q_q(|W|) \leq (t^2 + \sigma^2)^{1/2}$ if $E(W) = 0$. 105

The proof in the Appendix combines the fact that a log-concave density at its mean is at least e^{-1} times its maximum (Frédérizi, 1997; Frédérizi et al., 2016) with the heat equation for quantiles under Gaussian smoothing (Steinbrecher & Shaw, 2008), which give $d\{r(s) - B\}^2/ds \geq -1$ for $r(s) = Q_q(|W + s^{1/2}Z|)$. Unlike the expansion for a fixed smooth law, the bound is uniform over the class. The order x cannot be improved, even for mean-zero laws. 110

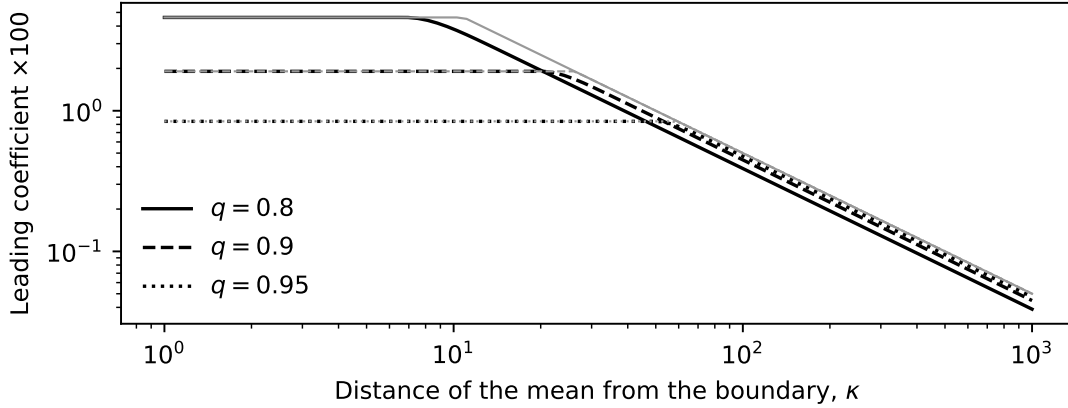


Fig. 1. Leading coefficient $L_q(\kappa)$ of the worst-case shortfall, times 100, when the mean lies at distance at least $\kappa x^{1/2}$ from the boundary (Theorem 3), for $q = 0.8$ (solid), 0.9 (dashes) and 0.95 (dots), with the upper bound $\min\{c_q, (\kappa^2 + 1)^{1/2} - \kappa\}$ from Theorems 1 and 2 (grey). Both axes are logarithmic.

Alt text: Line plot on logarithmic axes of the leading coefficient against the scaled distance of the mean from the boundary for three coverage levels. Each curve is flat at its unrestricted constant up to a threshold near 7, 20 and 50 and then decreases in proportion to the reciprocal of the distance, slightly below a grey upper bound.

PROPOSITION 2. *Let X have density proportional to $e^{\beta X}$ on $[0, \ell]$ with $\beta > 0$ and $\ell \leq \infty$, let $W = X - E(X)$ and let $r_0 = Q_q(|W|)$. If $\pm r_0$ lie in the interior of the support of W , then, after scaling W so that the noisy q -quantile of $|W + x^{1/2}Z|$ equals 1,*

$$Q_q(|W|) = 1 + \frac{1}{2}\beta r_0 \tanh(\beta r_0)x + o(x) \quad (x \rightarrow 0).$$

115 For every $q \in (0, 1)$ the condition holds for small $\beta\ell$, so $R_{q,q}^{(0)}(x) - 1$ is of exact order x .

The best coefficient in this family is 0.070 at $q = 0.9$, at $\beta\ell = 0.9$, and 0.347 at $q = 0.8$, for $W = 1 - E_1$; with Theorem 2, the worst-case coefficient over centred laws lies in $[0.070, 0.5]$ and $[0.347, 0.5]$ respectively. The widening is confined to small x ; once the noise reaches the end of the support the threshold may be shrunk.

120 3.4. The transition

THEOREM 3. *Let $1 - e^{-1} < q < 1$, $x > 0$ and $W \in \mathcal{L}$ with $\text{pr}(|W + x^{1/2}Z| \leq 1) \geq q$. Then $|E(W)| < 1$, and if $\delta = Q_q(|W|) - 1 > 0$,*

$$2\{1 - |E(W)|\}\delta + \delta^2 \leq x. \quad (3)$$

Moreover, for every $\kappa \geq 0$,

$$\lim_{x \rightarrow 0} x^{-1/2} \{R_{q,q}^{(b_x)}(x) - 1\} = L_q(\kappa), \quad b_x = 1 - \kappa x^{1/2}.$$

125 By (3), a shortfall $\delta \geq ax^{1/2}$ with $0 < a < 1$ forces $1 - |E(W)| \leq (1 - a^2)x^{1/2}/(2a)$, and the limit identifies the coefficient on that scale. In the original units the mean lies $\kappa D^{1/2}$ from the threshold, and the unrestricted worst case $c_q D^{1/2}$ persists until κ exceeds κ_q^* , about 20 noise standard deviations at $q = 0.9$; beyond it the coefficient decays like $\{1 - \log(1/q)\}/(2\kappa)$ (Fig. 1). The limit is taken with κ fixed and does not determine the coefficient for centred laws.

4. CALIBRATION SLACK AND A FINITE-SAMPLE PROCEDURE

4.1. Slack removes the shortfall

The proof of Theorem 1 splits the noisy failure budget between the two ends, and it extends to $p \geq q$:

$$R_{p,q}(x) \leq 1 + C_{p,q}x^{1/2}, \quad C_{p,q} = \sup_{\alpha_L + \alpha_R = 1-p} \inf_{\beta_L + \beta_R = 1-q} \max(c_{1-\alpha_L, 1-\beta_L}, c_{1-\alpha_R, 1-\beta_R}),$$

for every x . Numerically the worst split puts the whole budget on one end, so that $C_{p,q} = c_{p,q}$.

COROLLARY 1. *If $C_{p,q} \leq 0$ then $R_{p,q}(x) \leq 1$ for every x , and if $c_{p,q} > 0$ then $R_{p,q}(x) > 1$ for all small x . The smallest slack $p - q$ with $C_{p,q} \leq 0$ lies in $(4.3 \times 10^{-3}, 5.0 \times 10^{-3}]$, $(7.9 \times 10^{-4}, 1.0 \times 10^{-3}]$ and $(1.5 \times 10^{-4}, 2.0 \times 10^{-4}]$ at $q = 0.8, 0.9$ and 0.95 , and so does the smallest slack with $R_{q+\delta,q}(x) \leq 1$ for all x .*

With $K = 110$ calibration scores and rank $k = 105$, the order statistic certifies $p_k = 0.9036$ with probability 0.96, and $-0.0572 \leq C_{0.9036, 0.9} \leq -0.057$, so with homogeneous noise the noisy threshold is shrunk by at least the factor $1 - 0.057x^{1/2}$. The small-noise bound and the slack result need only concavity of $\log F$ and $\log(1 - F)$, so they hold for the possibly multimodal bi-log-concave laws of Dümbgen et al. (2017) (Supplementary Material).

4.2. Certified values

Between the small-noise regime and the edge of feasibility, $R_{p,q}$ is computed from Proposition 1. Members of $\mathcal{F}$ are stochastically monotone in their parameters, so a branch and bound over boxes of parameters certifies $R_{p,q}(x) \leq s$, and $R_{p,q}(x + h) \leq R_{p,q}(x) + c_q h^{1/2}$ fills the gaps between grid points. The closed forms agree with high-precision quadrature to 10^{-11} and boxes are cleared with a margin of 10^{-9} , which is not interval arithmetic. For $0.05 \leq x \leq 0.3$ the certified bound exceeds a feasible value by at most a relative 3×10^{-4} , and at $q = 0.9$ it falls below 1 from $x = 0.013$ on (Supplementary Material).

4.3. Heterogeneous and estimated noise variances

Let $W_1, \dots, W_K, W_{\text{new}}$ be independent with law $G \in \mathcal{L}$, independent of $e_i \sim N(0, D_i)$ with the D_i fixed and known. The scores $V_i = W_i + e_i$ are independent but not identically distributed.

PROPOSITION 3. *Let $T = |V|_{(k)}$, let $\bar{F}$ be the average distribution function of the $|V_i|$, assumed continuous, and let p_k be the δ -quantile of the $\text{Beta}(k, K + 1 - k)$ distribution. If $k \geq Kp_k + 1$ then $\text{pr}\{\bar{F}(T) \geq p_k\} \geq 1 - \delta$.*

The proof, by Hoeffding (1956), is in the Supplementary Material; the result is a case of the comparison of Sen (1970). On its event, G satisfies the single linear constraint $\int \bar{g} dG \geq p_k$ with the average kernel $\bar{g}(w) = K^{-1} \sum_i \text{pr}(|w + e_i| \leq T)$. Proposition 1 and the box bounds apply to $\bar{g}$, and the heterogeneous rule is the certified radius $s = TR_{p_k - \varepsilon, q}^{\text{mix}}(D_1/T^2, \dots, D_K/T^2)$, where $\varepsilon \approx 2 \times 10^{-4}$ bounds the error of compressing the D_i to 32 support points. Then $\text{pr}_{\mathcal{D}}\{\text{pr}(|W_{\text{new}}| \leq s \mid \mathcal{D}) \geq q\} \geq 1 - \delta$ for every $G \in \mathcal{L}$, with $\mathcal{D}$ the calibration data. The kernel of a single variance, the mean or the smallest, cannot replace $\bar{g}$ (Supplementary Material). With variances estimated independently of the scores, the supremum of the radius over a confidence set for the variances keeps the guarantee at level $1 - \delta - \eta$; under a common-scale model $D_i = \sigma^2 a_i$ it retains 62–74% of the gain over noisy conformal calibration in simulations (Supplementary Material).

Table 1. *Reliability, times 100, and mean width, in parentheses, of five interval rules; target reliability 95*

latent law	shape-free	Fay–Herriot	noisy conformal	mean variance	heterogeneous
normal	100 (5.92)	96.0 (3.93)	100 (4.96)	98.0 (4.46)	98.0 (4.45)
Laplace	100 (6.29)	92.7 (3.96)	100 (5.15)	98.7 (4.68)	98.7 (4.66)
centred gamma	100 (6.16)	100 (3.95)	100 (4.97)	100 (4.46)	100 (4.45)
truncated Laplace	100 (5.89)	84.7 (3.89)	100 (4.93)	97.3 (4.43)	96.7 (4.42)

The centred gamma law has shape parameter 2. Standard errors are at most 1.8 for reliabilities and 0.07 for widths, which are computed on the same data sets.

5. NUMERICAL ILLUSTRATION

The target is $\text{pr}_{\mathcal{D}}\{\text{pr}(W_{\text{new}} \in C \mid \mathcal{D}) \geq 0.9\} \geq 0.95$, with $K = 110$ calibration areas, $\text{var}(W) = 1$ and heterogeneous known noise variances $D_i = 0.577 L_i / E(L_i)$, L_i lognormal with log-scale 0.7. Conditional coverage is computed exactly from closed-form latent distribution functions, with 150 data sets per law. We compare, on the same data sets, the shape-free rule obtained from Markov’s inequality applied to $\bar{g}_T(0) - \bar{g}_T(W) \geq 0$, in the spirit of LatentCP, which for a fixed rank is the smallest radius the single constraint allows without a shape assumption, at rank 107, the best of 106–110 on these data; the Fay–Herriot tolerance interval with a parametric bootstrap; noisy split conformal calibration at rank 105; the log-concave rule with the mean variance, which has no guarantee under heterogeneous noise; and the certified heterogeneous rule of § 4.3.

Every heterogeneous radius was certified, at a cost of 0.2% of width, and the rule meets the target for all four laws. It is 9.5–10.5% shorter than noisy conformal calibration, which has no latent guarantee without a shape assumption, and 25–28% shorter than the shape-free rule, the price of making no shape assumption with this constraint. It is 12.5–18% longer than the Fay–Herriot interval, which misses the target under the truncated Laplace law. It is only 0.3% shorter than the mean-variance rule, so the exact kernel buys validity rather than width. The gains are in the cost of a guarantee, not in the interval shape: with widths rescaled by an oracle to equal reliability, noisy conformal calibration is not longer than the log-concave rules. An application to California school districts (Lumley, 2004) is in the Supplementary Material.

6. DISCUSSION

The guarantee is conditional on the calibration sample and concerns a new area whose residual is drawn from the law of the calibration residuals, not each fixed area or a fixed finite population; it assumes log-concave residuals and independent Gaussian noise with known variances. The mean condition of Theorems 2 and 3 concerns the latent residual law, which an intercept in the fitted regression does not centre.

Several questions remain. The leading coefficient for centred laws, between 0.070 and 0.5 at $q = 0.9$, is open, as are the analogue of Theorem 2 under a median condition, which Gaussian smoothing does not preserve, and a proof that one end carries the worst case, $C_{p,q} = c_{p,q}$. Variance models with a few parameters, such as generalized variance functions, are the natural next step for survey applications.

DECLARATION OF THE USE OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work the author used ChatGPT (OpenAI) and Claude (Anthropic) in order to check derivations, write and verify code, run numerical checks and edit the text.

After using these tools the author reviewed and edited the content as necessary and takes full responsibility for the content of the publication.

SUPPLEMENTARY MATERIAL

The Supplementary Material includes omitted proofs, the certificates, the estimated-variance procedure, further simulations and an application. Code reproducing every number is at <https://github.com/kunhailP/unsampled-area-intervals>.

APPENDIX 1

Proofs for § 3

The remaining proofs are in the Supplementary Material.

Proof of Lemma 2. Let $a = \log(1/q)$ and $A = 1 - a > 0$. Let Y be nondegenerate with q -quantile v . Since $\log F_Y$ is concave (Prékopa, 1973), a supergradient $u > 0$ at v gives $F_Y(y) \leq \min\{qe^{u(y-v)}, 1\}$, the distribution function of $Y' = v + a/u - E_u$; so Y' lies stochastically below Y and has the same q -quantile. Then $E(Y') \leq E(Y) \leq -\kappa$ and $\text{pr}(Y' + Z \leq 0) \geq q$, so the replacement keeps both constraints. For the tail $b - E_u$ the constraints read $b \leq b_u(q)$ and $b \leq u^{-1} - \kappa$, so the largest quantile is $\min\{v_q(u), A/u - \kappa\}$. If $k_q(u) < \kappa$ this minimum is $A/u - \kappa$; since k_q is continuous and $k_q(s) \rightarrow \infty$ as $s \rightarrow 0$, some $s < u$ has $k_q(s) = \kappa$, where the value is $A/s - \kappa > A/u - \kappa$. So only u with $k_q(u) \geq \kappa$ matter, and there the value is $v_q(u)$. Point masses satisfying the constraints are at most $-\Phi^{-1}(q) < 0 < L_q(\kappa)$, where positivity holds because $v_q(u) \sim u/2$ and $k_q(u) \sim A/u$ as $u \rightarrow 0$. The same expansions give $\kappa L_q(\kappa) \rightarrow A/2$: for large κ only small u are feasible, $u \leq \{A + o(1)\}/\kappa$, and $v_q(u) \leq \{1 + o(1)\}u/2$ with equality at the boundary. Monotonicity is clear, and $L_q(\kappa) = c_q$ while a maximizer of v_q is feasible. $\square$

Proof of Theorem 2. Let $\mu = E(W)$. If W is a point mass, $Q_q(|W|) = |\mu| \leq B$. Otherwise, for $s > 0$ the density f_s of $W + s^{1/2}Z$ is positive, smooth and log-concave with mean μ . Let $\ell_s = \log f_s$. By the mean-point inequality $\sup f_s \leq e f_s(\mu)$ (Frédérizi, 1997; Frédérizi et al., 2016, Corollary 4.6), $\ell_s(y) - \ell_s(\mu) \leq 1$ for all y , so concavity gives, for $r > B$,

$$\ell'_s(r) \leq \frac{\ell_s(r) - \ell_s(\mu)}{r - \mu} \leq \frac{1}{r - B}, \quad \ell'_s(-r) \geq \frac{\ell_s(\mu) - \ell_s(-r)}{\mu + r} \geq -\frac{1}{r - B}.$$

Let $H_s(r) = \text{pr}(|W + s^{1/2}Z| \leq r)$. By the heat equation $\partial_s f_s = f_s''/2$,

$$\partial_s H_s(r) = \frac{1}{2} \{f'_s(r) - f'_s(-r)\} \leq \frac{f_s(r) + f_s(-r)}{2(r - B)} = \frac{\partial_r H_s(r)}{2(r - B)}.$$

The quantile $r(s) = Q_q(|W + s^{1/2}Z|)$ solves $H_s\{r(s)\} = q$ and is continuously differentiable on $s > 0$, with $r'(s) \geq -1/[2\{r(s) - B\}]$ where $r(s) > B$. Hence $\{r(s) - B\}_+^2$ has derivative at least -1 almost everywhere, and $\{r(\epsilon) - B\}_+^2 \leq \{r(\sigma^2) - B\}_+^2 + \sigma^2 - \epsilon$. As $\epsilon \rightarrow 0$, $r(\epsilon) \rightarrow Q_q(|W|)$, because μ lies in the interior of the support of W , so that the distribution function of $|W|$ is continuous and strictly increasing below its upper end. Since $r(\sigma^2) \leq t$, the first inequality follows, and the second from $(d^2 + \sigma^2)^{1/2} \leq d + \sigma^2/(2d)$. $\square$

Proof of Proposition 2. Near $\pm r_0$ the density f of W is smooth with $f' = \beta f$, and the ends of the support are at positive distance. Hence $H_v(r) = \text{pr}(|W + v^{1/2}Z| \leq r) = H_0(r) + v\{f'(r) - f'(-r)\}/2 + o(v)$ uniformly near r_0 , and the noisy quantile is $t(v) = r_0 - v\beta\{f(r_0) - f(-r_0)\}/[2\{f(r_0) + f(-r_0)\}] + o(v) = r_0 - (\beta v/2) \tanh(\beta r_0) + o(v)$. Scaling by $t(v)$ and writing $x = v/t(v)^2$, a continuous and increasing function of v near 0, gives the expansion. As $\beta\ell \rightarrow 0$, W/ℓ tends to the uniform law on $[-1/2, 1/2]$, whose q -quantile of $|W|/\ell$ is $q/2 < 1/2$. $\square$

Proof of Theorem 3. Let $V = W + x^{1/2}Z$ with distribution function F . Since $\log F$ and $\log(1 - F)$ are concave and $F(V)$ is uniform, $E\{\log F(V)\} = -1$ and Jensen's inequality gives $F\{E(V)\} \geq e^{-1}$ and

likewise $1 - F\{E(W)\} \geq e^{-1}$. If $E(W) \geq 1$ then $\text{pr}\{|V| \leq 1\} \leq F\{E(W)\} \leq 1 - e^{-1} < q$, and similarly
 240 if $E(W) \leq -1$; so $|E(W)| < 1$. Theorem 2 with $B = |E(W)|$ and $t = 1$ gives $\delta \leq \{d^2 + x\}^{1/2} - d$ with
 $d = 1 - |E(W)|$, which is (3).

For the lower bound in the limit, take u with $k_q(u) \geq \kappa$ and the law $W = 1 + x^{1/2}\{b_u(q + \eta_x) - E_u\}$ of
 the lower bound in Theorem 1. Its mean is $1 + x^{1/2}\{b_u(q + \eta_x) - u^{-1}\} \leq 1 - \kappa x^{1/2}$, because b_u decreases
 in its argument, and it is positive for small x ; hence $\liminf_{x \rightarrow 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} \geq v_q(u)$ for every
 245 such u .

For the upper bound, suppose that $x_n \rightarrow 0$, $h_n = x_n^{1/2}$, and that W_n is feasible for $R_{q,q}^{(b_{x_n})}(x_n)$
 with $Q_q(|W_n|) - 1 \geq a_0 h_n$, where $L_q(\kappa) < a_0 < 1$. By (3), $\kappa h_n \leq 1 - |E(W_n)| \leq K h_n$ with $K = (1 -$
 $a_0^2)/(2a_0)$. Replacing W_n by $-W_n$ if necessary, $E(W_n) \geq 0$; let $Y_n = (W_n - 1)/h_n$, so that $-K \leq E(Y_n) \leq$
 $-\kappa$. The law of $T_n = Y_n + Z$ has a smooth log-concave density with maximum M_n and distribution
 250 function F_n , and $F_n(0) \geq \text{pr}\{|W_n + h_n Z| \leq 1\} \geq q$ while $F_n\{E(T_n)\} \leq 1 - e^{-1}$ by the first part. So
 $q - 1 + e^{-1} \leq F_n(0) - F_n\{E(T_n)\} \leq M_n K$. For a log-concave density g with maximum M and median
 m , the function $g\{F^{-1}(\cdot)\}$ is concave on $(0, 1)$ (Bobkov, 1996), so $g(m) \geq M/2$; the tangent bounds for
 $\log F$ and $\log(1 - F)$ at m give $\text{pr}\{|T - m| > y\} \leq e^{-2g(m)y}$, hence $\text{var}(T) \leq 1/\{2g(m)^2\} \leq 2/M^2$. Thus
 $\text{var}(Y_n) \leq 2K^2/(q - 1 + e^{-1})^2$, the Y_n are tight and uniformly integrable, and along a subsequence $Y_n \rightarrow Y$
 255 in distribution with Y log-concave, possibly a point mass, $E(Y) \leq -\kappa$ and $\text{pr}(Y + Z \leq 0) = E\{\Phi(-Y)\} \geq q$.
 On $\{W_n \geq 0\}$, whose probability tends to one, $(|W_n| - 1)/h_n = Y_n$, so $a_0 \leq \{Q_q(|W_n|) - 1\}/h_n \rightarrow$
 $Q_q(Y) \leq L_q(\kappa)$ by Lemma 2, where the quantile converges because a log-concave law has a unique
 q -quantile or is a point mass. This contradicts $a_0 > L_q(\kappa)$, so $\limsup_{x \rightarrow 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} \leq a_0$ for
 every $a_0 \in (L_q(\kappa), 1)$, and $L_q(\kappa) \leq c_q < 1$. $\square$

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